

SINAH LEGNER, MSc.

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Education

University of Cambridge

Oct. 2022 – Present

PhD in Physics

- Currently pursuing PhD under the supervision of Dr. Will Handley and Dr. Will Barker, focusing on testing the Poincaré Gauge Theory of gravity through exact solutions at both cosmological and stellar scales.

ETH Zurich

2019 – 2022

MSc. Physics

- Relevant courses: QFT I (canonical quantization), QFT II (path integral formulation), Advanced Field Theory, Symmetries in Physics, Astrophysics, and Theoretical Cosmology and Different Aspects of Gravity.
- Semester project on line shape calculation of muonium Lamb shift, focusing on simulating uncertainties using Monte-Carlo method.
- Master's thesis with Professor Lavinia Heisenberg on propagating gravitational waves through cosmic structure for scalar-tensor and vector-tensor theories of gravity.

King's College London

2016 – 2019

BSc. Physics with Theoretical Physics

- First-class honours degree covering quantum mechanics, solid state physics, general relativity, and advanced mathematical methods including complex analysis.
- Third year literature review on quartz crystal oscillators.

Shanghai High School International Division

2008 – 2015

High School Diploma

- AP (Advanced Placement) Scholar with Distinction award with AP scores of 5/5 for Physics (1, 2, C:Mechanics, and C:Electricity and Magnetism), Chemistry, Biology, Statistics, and Chinese Language and Culture.

Publications

First Author

- Sinah Legner, Will Handley, Will Barker, and Adam Ormondroyd, 'Alleviating the Hubble Tension with Torsion Condensation', arXiv preprint: 2507.09228 [astro-ph.CO]

Work Experience

University of Cambridge

2022

Undergraduate Supervisor – General Relativity

- Supervised three groups of four third year undergraduate students during the first term of PhD, grading homework and conducting biweekly hour-long problem-solving sessions.

King's College London

2019

Undergraduate Tutor – Nuclear Physics

- Part-time tutoring of second year students in nuclear physics while completing full-time undergraduate degree, developing time management and communication skills.

King's College London

2018

Research Intern

- 7-week research internship in Professor Malcolm Fairbairn's group, developing Python programs to fit multi-variable equations to observational data using Chi-square tests. Continued part-time work on the project during undergraduate studies.

Conferences and Workshops

Research Visits

- September 2025, Short-Term Scientific Mission (STSM), University of Tartu, Estonia – Collaborated on implementing the Hyperdust model into CAMB Boltzmann code for testing with DESI data. Led workshop on CAMB and Cobaya data analysis framework. (COST Action CA21136)

Talks

- September 2025, ‘Alleviating the Hubble Tension with Torsion Condensation’, University of Tartu, Estonia
- July 2025, Presented research overview and PhD opportunities to visiting undergraduates from China, Cavendish Laboratory, University of Cambridge
- February 2025, ‘The Capacity of Constant Torsion Emergent Gravity to Resolve Cosmological Tensions’, Hill’s Coffee Talk, Battcock Centre, University of Cambridge

Posters

- 2025, ‘Alleviating the Hubble Tension with Torsion Condensation’, GR24 and Amaldi16, Scottish Exhibition Centre
- 2025, ‘Alleviating the Hubble Tension with Torsion Condensation’, CosmoVerse@Istanbul, Istanbul Technical University
- 2025, ‘Constant Torsion Emergent Gravity vs Cosmological Tensions’, Cosmological Frontiers in Fundamental Physics, Université Paris Cité
- 2024, ‘Poincare Gauge Theory versus Cosmological Tension’, CosmoVerse@Krakow, Jagiellonian University

Other Workshops

- 2019, ‘Global Poverty, Human Rights, and Development’, Helsinki University – Interdisciplinary course on poverty, human rights, and development featuring case studies, debates, and presentations.

Extracurricular

Cambridge University Women in Physics Society

2023 – 2026

Publicity Officer → *Chair* → *Secretary*

- **Secretary (2025–2026):** Managing society administration and communications.
- **Chair (2024–2025):** Leadership role organizing academic and social events including PhD application panels, internship panels, lunch socials, garden parties, and welfare teas. Collaborated with the Cavendish Laboratory to organize department-wide LGBTQ+ Day and International Women’s Day celebrations.
- **Publicity Officer (2023–2024):** Created promotional materials, publicized events, and organized speaker sessions. Led International Women’s Day panel session for undergraduates at the Institute of Astronomy.

Cambridge University Revolver and Pistol Club

2024 – 2026

Communications Officer → *Webmaster*

- **Webmaster (2025–2026):** Managing club website and digital communications.
- **Communications Officer (2024–2025):** Supported club members with shooting-related questions and completed training as a range conducting officer for safety and security operations.
- Won the 2025 Gallery Rifle Varsity Match as a member of the university team.

Technical Skills

Languages: Chinese (native), English (fluent)

Programming: Python (Intermediate), Mathematica (Intermediate), L^AT_EX (Intermediate), Julia (Beginner), Fortran (Beginner), Command-Line Tools and Text Editing (Intermediate)

Other Skills: Graphic design, oil painting